Name
Date
Course

Applications of Algebra (Part II): Solving Equations

Applications of One-Step Equations

① If the area of a rectangle is 30 cm², and the length is 5 cm, what is the width?

$$A = LW$$

$$30 = 5W$$

$$\frac{30}{5} = \frac{5W}{5}$$

$$6 = W$$

$$\boxed{6\text{ cm}}$$

② If the height and area of a parallelogram are 20 feet and 10 square feet, what is the base?

$$A = bh$$

$$10 = b(20)$$

$$\frac{10}{20} = \frac{b(20)}{20}$$

$$\boxed{\tfrac{1}{2}\text{ ft} = b}$$

③ If the area of a rectangle is 40.5 m², and the length is 9 m, what is the width?

$$A = LW$$

$$40.5 = 9W$$

$$\frac{40.5}{9} = \frac{9W}{9}$$

$$\boxed{4.5\text{ m} = W}$$

④ If the area and width of a rectangle are 0.8 in² and 1.8 in, what is the length?

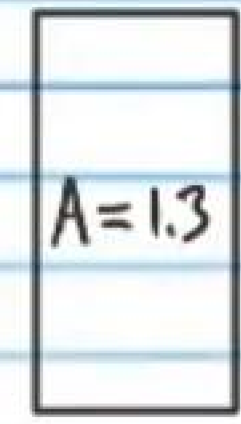
$$A = LW$$

$$0.8 = L(1.8)$$

$$\frac{0.8}{1.8} = \frac{L(1.8)}{1.8}$$

$$\boxed{0.\bar{4}\text{ in} = L}$$

⑤



L = 20
W = ?

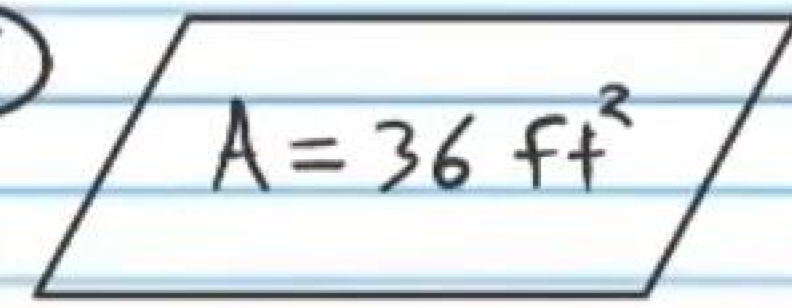
$$A = LW$$

$$1.3 = 20W$$

$$\frac{1.3}{20} = \frac{20W}{20}$$

$$\boxed{0.065 = W}$$

⑥



b = 4 ft
h = ?

$$A = bh$$

$$36 = 4h$$

$$\frac{36}{4} = \frac{4h}{4}$$

$$\boxed{9\text{ ft} = h}$$

⑦ r = 4 mi/h
t = 7 h
d = ?

$$r = \frac{d}{t}$$

$$4 = \frac{d}{7}$$

$$(7)4 = \frac{d}{7}(7)$$

$$\boxed{28\text{ mi} = d}$$

⑧ r = 8.4 m/s
t = 3 s
d = ?

$$r = \frac{d}{t}$$

$$8.4 = \frac{d}{3}$$

$$(3)8.4 = \frac{d}{3}(3)$$

$$\boxed{25.2\text{ m} = d}$$

⑨ If a car travels 64 mi/h for 5 hours how far will it move?

$$r = \frac{d}{t}$$

$$64 = \frac{d}{5}$$

$$(5)64 = \frac{d}{5}(5)$$

$$\boxed{320\text{ mi} = d}$$

Finding a Percent of a Number

Example 1: 79% of 21.

$$0.79 \times 21 = \boxed{16.59}$$

Example 2: 8.2% of 56.

$$0.082 \times 56 = \boxed{4.592}$$

Example 3: 233% of 19.

$$2.33(19) = \boxed{44.27}$$

(10) If 23% of a number is 12.88, what is the number?

$$0.23X = 12.88$$

$$\frac{0.23X}{0.23} = \frac{12.88}{0.23}$$

$$\boxed{X = 56}$$

(11) 4.2 percent of what number is 0.378?

$$0.042X = 0.378$$

$$\frac{0.042X}{0.042} = \frac{0.378}{0.042}$$

$$\boxed{X = 9}$$

(12) If 83% of a number is 1.494, what is the number?

$$0.83X = 1.494$$

$$\frac{0.83X}{0.83} = \frac{1.494}{0.83}$$

$$\boxed{X = 1.8}$$

(13) 160% of what number is 128?

$$1.6X = 128$$

$$\frac{1.6X}{1.6} = \frac{128}{1.6}$$

$$\boxed{X = 80}$$

(14) If 0.0003 percent of a number is 0.0000156, what is the number?

$$0.000003X = 0.0000156$$

$$\frac{0.000003X}{0.000003} = \frac{0.0000156}{0.000003}$$

$$\boxed{X = 5.2}$$

(15) 49% of what number is 87.22?

$$0.49X = 87.22$$

$$\frac{0.49X}{0.49} = \frac{87.22}{0.49}$$

$$\boxed{X = 178}$$

16 What percent of 96 is 7.68?

$$X(96) = 7.68$$

$$\frac{X(96)}{96} = \frac{7.68}{96}$$

$$X = 0.08$$

$$\boxed{8\%}$$

17 25.5 is what percent of 102?

$$25.5 = 102X$$

$$\frac{25.5}{102} = \frac{102X}{102}$$

$$0.25 = X$$

$$\boxed{25\%}$$

18 What percent of 8 is 0.24?

$$8X = 0.24$$

$$\frac{8X}{8} = \frac{0.24}{8}$$

$$X = 0.03$$

$$\boxed{3\%}$$

19 60 is what percent of 54.6?

$$60 = 54.6X$$

$$\frac{60}{54.6} = \frac{54.6X}{54.6}$$

$$1.0989 \approx X$$

$$\boxed{109.89\%}$$

20 What percent of 29 is 58.87?

$$29X = 58.87$$

$$\frac{29X}{29} = \frac{58.87}{29}$$

$$X = 2.03$$

$$\boxed{203\%}$$

21 10.2 is what percent of 85?

$$10.2 = 85X$$

$$\frac{10.2}{85} = \frac{85X}{85}$$

$$0.12 = X$$

$$\boxed{12\%}$$

Applications of Two-Step Equations

(22) If the perimeter and width of a rectangle are 26 cm and 10 cm respectively, then what is the length of the rectangle?

$$P = 2W + 2L$$
$$26 = 2(10) + 2L$$
$$26 = 20 + 2L$$
$$-20 + 26 = 20 + 2L - 20$$
$$6 = 2L$$
$$\frac{6}{2} = \frac{2L}{2}$$
$$\boxed{3\text{cm} = L}$$

(23)

$P = 70$ in
$L = 15$ in
$W = ?$

$$P = 2W + 2L$$
$$70 = 2W + 2(15)$$
$$70 = 2W + 30$$
$$-30 + 70 = 2W + 30 - 30$$
$$40 = 2W$$
$$\frac{40}{2} = \frac{2W}{2}$$
$$\boxed{20\text{in} = W}$$

(24) If the perimeter of a rectangle is 34 feet, and the length is 10 feet, what is the width?

$$P = 2W + 2L$$
$$34 = 2W + 2(10)$$
$$34 = 2W + 20$$
$$-20 + 34 = 2W + 20 - 20$$
$$14 = 2W$$
$$\frac{14}{2} = \frac{2W}{2}$$
$$\boxed{7\text{ft} = W}$$

(25)

$P = 3.8$
$W = 0.45$
$L = ?$

$$P = 2W + 2L$$
$$3.8 = 2(0.45) + 2L$$
$$3.8 = 0.9 + 2L$$
$$-0.9 + 3.8 = 0.9 + 2L - 0.9$$
$$2.9 = 2L$$
$$\frac{2.9}{2} = \frac{2L}{2}$$
$$\boxed{1.45 = L}$$

Percent Change

$$N = F \pm Fr$$

Example 1: If 63 is increased by 72%, what does it become?

$$N = 63 + 0.72(63)$$
$$= 63 + 45.36 = \boxed{108.36}$$

Example 2: Bill's score on his new video game was 170. If he increased his score by 20% the second time he played, what was his new score?

$$N = F + Fr = 170 + 0.2(170)$$
$$170 + 34 = \boxed{204}$$

Example 3: Arnold's weight decreased by 5%. If his previous weight was 185 pounds, what is his new weight?

$$N = F - Fr = 185 - 0.05(185)$$
$$185 - 9.25 = \boxed{175.75\text{ lbs}}$$

Example 4: Linda bought a motorcycle. Her savings decreased by 71%. Before she bought the motorcycle, her total savings were \$11,123. How much money does she have in savings now?

$$N = F - Fr = 11{,}123 - (0.71)11{,}123$$
$$11{,}123 - 7{,}897.33 = \boxed{\$3{,}225.67}$$

(26) If a number is decreased from 75 to 66, what is the percent decrease?

$$N = F - Fr$$
$$66 = 75 - 75r$$
$$-75 + 66 = 75 - 75r - 75$$
$$-9 = -75r$$
$$0.12 = r$$
$$\boxed{12\%}$$

(27) If a number is increased from 20 to 56, what is the percent increase?

$$N = F + Fr$$
$$56 = 20 + 20r$$
$$-20 + 56 = 20 + 20r - 20$$
$$36 = 20r$$
$$\frac{36}{20} = \frac{20r}{20}$$
$$1.8 = r$$
$$\boxed{180\%}$$

(28) After modifying his recipe, a chef's cookies were reduced in volume from 0.5 ounces to 0.1 ounces. What is the percent decrease of these cookies?

$$N = F - Fr$$
$$0.1 = 0.5 - 0.5r$$
$$-0.5 + 0.1 = 0.5 - 0.5r - 0.5$$
$$-0.4 = -0.5r$$
$$\frac{-0.4}{-0.5} = \frac{-0.5r}{-0.5}$$
$$0.8 = r$$
$$\boxed{80\%}$$

(29) Jared increased the area of his posters from 5.5 ft^2 to 6 ft^2. Calculate the percent increase of the area of the posters.

$$N = F + Fr$$
$$6 = 5.5 + 5.5r$$
$$-5.5 + 6 = 5.5 + 5.5r - 5.5$$
$$0.5 = 5.5r$$
$$\frac{0.5}{5.5} = \frac{5.5r}{5.5}$$
$$0.\overline{09} = r$$
$$0.09090909$$
$$\boxed{9.\overline{09}\%}$$

(30) If 80 is decreased to 30, what is the percent decrease?

$$N = F - Fr$$
$$30 = 80 - 80r$$
$$-80 + 30 = 80 - 80r - 80$$
$$-50 = -80r$$
$$\frac{-50}{-80} = \frac{-80r}{-80}$$
$$0.625 = r$$
$$\boxed{62.5\%}$$

(31) If 5 is decreased to 3, what is the percent decrease?

$$N = F - Fr$$
$$3 = 5 - 5r$$
$$-5 + 3 = 5 - 5r - 5$$
$$-2 = -5r$$
$$\frac{-2}{-5} = \frac{-5r}{-5}$$
$$0.4 = r$$
$$\boxed{40\%}$$

★ (32) If a number is decreased by 96% and the result is 30, what was the original number?

$$N = F - Fr$$

$$30 = F - F(0.96)$$

$$30 = F - 0.96F$$

$$30 = 0.04F$$

$$\frac{30}{0.04} = \frac{0.04F}{0.04}$$

$$\boxed{750 = F}$$

(33) If the average length of a lizard is increased by 4%, and the new average length is 5.408 centimeters, what was the original length?

$$N = F + Fr$$

$$5.408 = F + F(0.04)$$

$$5.408 = F + 0.04F$$

$$5.408 = 1.04F$$

$$\frac{5.408}{1.04} = \frac{1.04F}{1.04}$$

$$\boxed{5.2\text{ cm} = F}$$

(34) If a number is increased 6% to 25.44, what was the original number?

$$N = F + Fr$$

$$25.44 = F + F(0.06)$$

$$25.44 = 1.06F$$

$$\frac{25.44}{1.06} = \frac{1.06F}{1.06}$$

$$\boxed{24 = F}$$

(35) If a baby elephant's weight increases 250% from the time it was born and its new weight is 770 lbs, what was his weight at birth?

$$N = F + Fr$$

$$770 = F + F(2.5)$$

$$770 = 3.5F$$

$$\frac{770}{3.5} = \frac{3.5F}{3.5}$$

$$\boxed{220\text{ lbs} = F}$$

(36) If a number is decreased 17% and the result is 0.0249, what was the original number?

$$N = F - Fr$$

$$0.0249 = F - F(0.17)$$

$$0.0249 = 0.83F$$

$$0.0249 = 0.83F$$

$$\boxed{0.03 = F}$$

(37) After dropping 7%, the new price of prime steak is $33.48 per pound. What was the original cost per pound?

$$N = F - Fr$$

$$33.48 = F - F(0.07)$$

$$33.48 = 0.93F$$

$$\frac{33.48}{0.93} = \frac{0.93F}{0.93}$$

$$\boxed{\$36 = F}$$

Applications of Three-Step Equations

(38) Stacey deposits $400 in a bank account which earns simple interest paid annually. After four years, she has $411.20 in the account. What was the interest rate of the account?

$$A = P + Prt$$

$$411.20 = 400 + 400r(4)$$

$$411.20 = 400 + 1600r$$

$$-400 + 411.20 = 400 + 1600r - 400$$

$$11.20 = 1600r$$

$$\frac{11.20}{1600} = \frac{1600r}{1600}$$

$$0.007 = r$$

$$\boxed{0.7\% = r}$$

(39) You invest $80 in a savings account that pays 6% simple interest every year. After a period of time the account grows to $113.60. How much time passed since you created the account?

$$A = P + Prt$$

$$113.60 = 80 + 80(0.06)t$$

$$113.6 = 80 + 4.8t$$

$$-80 + 113.6 = 80 + 4.8t - 80$$

$$33.6 = 4.8t$$

$$\frac{33.6}{4.8} = \frac{4.8t}{4.8}$$

$$\boxed{7 \text{ years} = t}$$

(40) You loan someone \$2009 at seven percent simple interest paid daily. After a certain amount of time, the person owes you \$7071.68. How much time passed?

$$A = P + Prt$$

$$7071.68 = 2009 + 2009(0.07)t$$

$$7071.68 = 2009 + 140.63t$$

$$-2009 + 7071.68 = 2009 + 140.63t - 2009$$

$$5062.68 = 140.63t$$

$$\frac{5062.68}{140.63} = \frac{140.63t}{140.63}$$

$$\boxed{36 \text{ days} = t}$$

(41) You take out a \$600 loan from a bank that charges 2.3% simple interest each month. After time goes by, you owe \$931.20. How much time passed?

$$A = P + Prt$$

$$931.2 = 600 + 600(.023)t$$

$$931.2 = 600 + 13.8t$$

$$-600 + 931.2 = 600 + 13.8t - 600$$

$$331.2 = 13.8t$$

$$\frac{331.2}{13.8} = \frac{13.8t}{13.8}$$

$$\boxed{24 \text{ months} = t}$$

Applications of Equations Requiring Four or More Steps

In the following problems, x represents the final position of an object, x_0 is its starting position, V_0 is its initial velocity, "a" is its acceleration, and t is the time passed since it started moving. Use the formula below and the given information to find the unknown.

$$x = x_0 + V_0 t + \frac{at^2}{2}$$

(42) $x = 31$, $x_0 = 20$, $t = 1$, $a = -2$, $V_0 = ?$

$$x = x_0 + V_0 t + \frac{at^2}{2}$$

$$31 = 20 + V_0(1) + \frac{-2(1)^2}{2}$$

$$31 = 20 + V_0 + (-1)$$

$$31 = 19 + V_0$$

$$-19 + 31 = 19 + V_0 - 19$$

$$\boxed{12 = V_0}$$

(43) $x = -15$, $x_0 = 10$, $V_0 = 2$, $t = 3$, $a = ?$

$$x = x_0 + V_0 t + \frac{at^2}{2}$$

$$-15 = 10 + 2(3) + \frac{a(3)^2}{2}$$

$$-15 = 10 + 6 + \frac{9a}{2}$$

$$-16 \quad -15 = 16 + \frac{9a}{2} - 16$$

$$-31 = \frac{9a}{2}$$

$$(2)(-31) = \frac{9a}{2}(2)$$

$$-62 = 9a$$

$$\frac{-62}{9} = \frac{9a}{9}$$

$$\boxed{-6.\overline{8} = a}$$

(44) $X=2,\ X_0=-1,\ t=2,\ a=3,\ V_0=?$

(45) $X=-1,\ X_0=1,\ t=4,\ V_0=2,\ a=?$

(46) The sum of three consecutive even integers is 252. Identify these integers.

$1^{st}=X$

$2^{nd}=X+2$

$3^{rd}=X+4$

$$X+(X+2)+(X+4)=252$$

$$X+X+2+X+4=252$$

$$3X+6=252$$

$$-6+3X+6=252-6$$

$$3X=246$$

$$\frac{3X}{3}=\frac{246}{3}$$

$$X=82$$

$$\boxed{82, 84, 86}$$

(47) The sum of four consecutive odd integers is 152. What are these integers?

$1^{st}=X$

$2^{nd}=X+2$

$3^{rd}=X+4$

$4^{th}=X+6$

$$X+(X+2)+(X+4)+(X+6)=152$$

$$X+X+2+X+4+X+6=152$$

$$4X+12=152$$

$$-12+4X+12=152-12$$

$$4X=140$$

$$\frac{4X}{4}=\frac{140}{4}$$

$$X=35$$

$$\boxed{35, 37, 39, 41}$$

(48) If the sum of three consecutive odd integers is 183, what are these integers?

$1^{st} = x$

$2^{nd} = x+2$

$3^{rd} = x+4$

$$x+(x+2)+(x+4)=183$$

$$3x+6 = 183$$

$$-6+3x+6 = 183-6$$

$$3x = 177$$

$$\frac{3x}{3} = \frac{177}{3}$$

$$x = 59$$

$$\boxed{59, 61, 63}$$

(49) If the sum of four consecutive even integers is 308, what are these integers?

$1^{st} = x$

$2^{nd} = x+2$

$3^{rd} = x+4$

$4^{th} = x+6$

$$x+(x+2)+(x+4)+(x+6)=308$$

$$4x+12 = 308$$

$$-12+4x+12 = 308-12$$

$$4x = 296$$

$$\frac{4x}{4} = \frac{296}{4}$$

$$x = 74$$

$$\boxed{74, 76, 78, 80}$$